

1. EXECUTIVE SUMMARY

Meridian Voice AI automates motor insurance First Notice of Loss (FNOL) intake, replacing 20-minute call center queues with sub-second (<800ms) conversational AI. The system pairs real-time voice streaming and single-pass LLM entity extraction with a deterministic state machine, delivering 100% regulatory compliance and non-blocking outbox persistence.

2. SYSTEM ARCHITECTURE



3. ARCHITECTURAL TRADE-OFFS

Design Choice	Strategic Benefit (Pros)	Architectural Trade-off (Cons)
Deterministic FSM vs Pure LLM	100% regulatory compliance & zero misrouted emergencies	Slightly reduced conversational flexibility
In-Memory State vs Redis	Sub-millisecond turn latency with zero network overhead	State tied to container lifecycle; single-pod bottleneck
Google Sheets vs Relational DB	Immediate, human-readable claim portal for adjusters	Subject to API rate limits (429) & lacks ACID guarantees
Flash LLM vs Frontier Model	Sub-350ms TTFT; 90% lower cost per claim turn	Slightly lower reasoning depth on complex edge cases
Async vs Sync Persistence	Decouples persistence I/O from real-time audio budget	Requires background fault mitigation for dropped tasks

4. KEY ARCHITECTURAL DECISIONS

- **Hybrid Orchestration:** Stochastic LLM handles entity extraction; deterministic FSM strictly enforces policy checks and medical escalations.
 - **Sub-Second Latency Budget:** Flash LLM over Server-Sent Events delivers ~350ms TTFT to meet <800ms glass-to-glass latency target.
- **Asynchronous Persistence:** Persistence writes execute out-of-band to ensure zero voice audio lag or turn blocking.
 - **Single-Pass Extraction:** Enforced JSON schema returns spoken language and slots in a single network turn.

5. FAILURE MODES & RESILIENCE MATRIX

Failure Mode	Business Impact	Current Mitigation	Production Solution
Container Restart	In-flight state reset	Session cleanup on disconnect	Redis Cluster + Redlock mutex
Sheets API Down	Storage delay	Local disk outbox buffer	Durable Kafka event streaming
LLM Gateway Timeout	Turn stall	Secondary API provider failover	Circuit breaker + multi-region
Email API Error	Delayed notification	Silent failure recovery log	Worker retry queue with backoff

6. ARCHITECTURAL EVOLUTION & 7. ARCHITECT'S PHILOSOPHY

PROTOTYPE PHASE

- In-Memory Session Map
- Single Container Instance
- Async Sheets + Disk Outbox

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PRODUCTION EVOLUTION

- Distributed Redis Cluster
- Stateless K8s Pod
- Auto-scaling
- Kafka + PostgreSQL + OTel

ARCHITECT'S PHILOSOPHY: This architecture establishes a strict separation between **stochastic intelligence** (LLM entity extraction) and the **deterministic control plane** (state machine business rules). By offloading heavy media processing to telephony gateways and executing persistence out-of-band, the system guarantees strict regulatory compliance while maintaining an enterprise-grade <800ms voice latency budget.